SOFTWARE TEST PLAN

Concurrent File Sharing System Using SDN

TEAM NUMBER: 19

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1. Test Plan Identifier

Project Title: Concurrent File Sharing System Using SDN

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Test Plan ID: T19-STP-001

Version: 1.0

Date: 12 November 2026

Prepared By: Team 19

Testing Level: Integration Testing and System Testing

Team Members:

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Project Repository: https://github.com/srushti-puranik/concurrent-file-sharing-system

Revision history is recorded in the Document Control table.

2. Introduction

The purpose of this Software Test Plan is to describe the testing activities for the Concurrent File Sharing System Using SDN developed by Team 19.

The project provides a platform where multiple clients can upload, download, and access files through a central server. TCP socket programming is used for client-server communication, while Software Defined Networking (SDN) is used to manage network traffic and enforce network policies.

This plan covers testing of file upload and download operations, user authentication, file access permissions, bandwidth management, client blocking and unblocking, and concurrent file transfers.

The objective is to verify that the system performs its main functions correctly, maintains file integrity, supports multiple clients, and applies the configured SDN policies.

References:

- Project source code and repository.
- Project Software Requirements Specification (SRS), where available.
- IEEE 829-1998 Test Plan format.

3. Test Items

No.	Test Item	Description
1	Client module	Handles client requests to upload, download, and list files.
2	Server module	Receives client requests and manages file storage and transfers.
3	User authentication	Verifies credentials and permits or rejects access.
4	File access control	Checks ownership and permissions before allowing downloads.
5	SDN controller	Manages network policies and communicates with network switches.
6	Bandwidth management	Applies configured bandwidth limits to clients.
7	Blocking and unblocking	Supports temporary blocking after repeated failed logins and manual policy control.
8	Concurrent file transfer	Exercises simultaneous file transfers by multiple clients.
9	Performance measurement	Records throughput, aggregate throughput, and fairness.

4. Features to Be Tested

No.	Feature	Expected Result	Priority
1	User login	Valid credentials are accepted and invalid credentials are rejected.	High
2	File upload	An authenticated user can upload a file successfully.	High
3	File listing	Uploaded files are displayed in the file list.	Medium
4	File download	An authorized user can download a file successfully.	High
5	File permissions	A user without permission is denied access to a restricted file.	High
6	File integrity	The downloaded file matches the original uploaded file.	High
7	Failed-login blocking	Three consecutive failed login attempts trigger a temporary block.	High
8	Manual blocking	The selected client can be blocked and unblocked by controller command.	High
9	Bandwidth throttling	The configured bandwidth limit is applied to the selected client.	High
10	Concurrent transfers	Multiple clients can perform transfers simultaneously.	High
11	Performance testing	Throughput and fairness measurements are collected.	Medium

5. Features Not to Be Tested

The following features are outside the scope of the current test cycle:

4. Operation over the public internet or large-scale production networks.
5. Compatibility with operating systems other than the configured Ubuntu environment.
6. Advanced graphical user-interface testing, as the current project primarily uses command-line interaction.
7. Encryption of stored files and passwords, as these are not implemented in the current version.
8. Recovery from physical hardware failures or complete server crashes.
9. Performance with an unlimited number of clients beyond the configured Mininet topology.

6. Approach

6.1 Testing Strategy

The system is tested using functional testing, integration testing, security testing, SDN policy testing, and performance testing.

6.2 Testing Methods

- Functional testing: Verify login, file upload, file listing, and file download.
- Integration testing: Check communication between the client, server, Mininet network, and SDN controller.
- Security testing: Verify invalid-login handling, file permissions, and temporary blocking after repeated failed login attempts.
- SDN policy testing: Verify manual blocking, unblocking, and bandwidth throttling.
- Performance testing: Measure file-transfer performance with different file sizes and numbers of simultaneous clients.

6.3 Testing Tools and Environment

Ubuntu Linux, Python 3, Mininet, POX controller, Open vSwitch (OVS), TCP socket programming, terminal tools, CSV result files, and Python graph-generation scripts are used.

6.4 Test Execution Procedure

10. Start the POX SDN controller.
11. Create and start the Mininet network topology.
12. Verify the required client-to-server communication.
13. Perform login tests using valid and invalid credentials.
14. Test file upload, listing, download, and access permissions.
15. Test temporary blocking after three consecutive failed login attempts.
16. Verify manual blocking, unblocking, and bandwidth throttling.
17. Perform concurrent transfers using different client counts and file sizes.

18. Record results and verify file integrity.
19. Generate performance graphs for analysis.

6.5 Performance Measurements

Measurements include mean throughput per client, aggregate throughput, minimum and maximum throughput, Jain's fairness index, and file-integrity verification.

The performance test matrix compares SDN-enabled and SDN-disabled runs using 5 MB and 20 MB files and 1, 3, 5, and 10 simultaneous clients.

6.6 Constraints

Testing is performed in the configured Mininet virtual network. Performance can vary with available system resources. ICMP ping traffic is restricted by the configured SDN policy; therefore, ping failures alone are not treated as file-transfer failures.

7. Item Pass/Fail Criteria

A test case is considered PASS when the actual result matches the expected result. It is considered FAIL when the expected functionality is not achieved. Failed cases must be recorded and investigated.

No.	Test Item	Pass Criteria
1	User authentication	Valid login succeeds and invalid login is rejected.
2	File upload	The file is uploaded successfully to the server.
3	File listing	The uploaded file appears in the file list.
4	File download	An authorized user downloads the file successfully.
5	File access control	Unauthorized access to restricted files is denied.
6	File integrity	The downloaded file matches the original.
7	Automatic blocking	The client is temporarily blocked after three consecutive failed login attempts.
8	Manual blocking	The selected client is blocked and subsequently unblocked.
9	Bandwidth throttling	The configured bandwidth limit is reflected in the client policy/status.
10	Concurrent transfers	Participating clients complete transfers and integrity checks succeed.
11	Performance testing	Measurements are recorded for the specified test matrix and integrity checks succeed.

All critical test cases must pass for the corresponding functionality to be considered successful. The execution status of each case is recorded separately in the test results.

8. Suspension Criteria and Resumption Requirements

8.1 Suspension Criteria

Testing will be temporarily suspended if:

20. The file server fails to start or becomes unavailable.
21. The SDN controller cannot communicate with the network switches.
22. The Mininet topology cannot be created successfully.
23. Critical errors prevent test execution.
24. The testing environment becomes unstable or unavailable.

8.2 Resumption Requirements

Testing may resume after:

25. The server and SDN controller are running correctly.
26. The Mininet topology has been recreated successfully.
27. Required network connections and services are restored.
28. Critical errors have been resolved.
29. Affected test cases have been rerun to confirm correct operation.

9. Test Deliverables

No.	Deliverable	Description
1	Software Test Plan	Document describing scope, strategy, environment, and schedule.
2	Test cases	Test scenarios with expected and actual results.
3	Test execution results	Records of passed, failed, and not-yet-executed test cases.
4	Test logs	Server, client, and SDN controller execution logs.
5	Functional test evidence	Screenshots or records of login, file operations, and permission tests.
6	SDN policy evidence	Evidence of blocking, unblocking, and bandwidth throttling.
7	Performance results	CSV files containing throughput, fairness, and integrity measurements.
8	Performance graphs	Graphs for per-client throughput, aggregate throughput, and fairness.
9	Final test summary	Summary of testing activities, results, and limitations.

Project result files:

- results/perf_raw.csv
- results/perf_summary.csv
- results/bandwidth_log.csv
- results/graph_per_client.png
- results/graph_aggregate.png
- results/graph_fairness.png

10. Test Tasks

No.	Test Task	Description
1	Test environment setup	Configure Ubuntu, Mininet, POX, and required Python packages.
2	Controller testing	Start the POX controller and verify switch communication.
3	Network setup	Create the Mininet topology containing the server and client hosts.
4	Authentication testing	Test valid and invalid user credentials.
5	File operation testing	Test upload, listing, download, and file integrity.
6	Permission testing	Verify that users cannot access files without permission.
7	Security testing	Test temporary blocking after repeated failed login attempts.
8	SDN policy testing	Verify manual blocking, unblocking, and bandwidth throttling.
9	Concurrent transfer testing	Perform simultaneous file transfers using multiple clients.
10	Performance testing	Run tests with SDN enabled and disabled.
11	Result analysis	Analyze CSV results, generate graphs, and prepare documentation.

Tasks are performed in sequence, beginning with environment setup and ending with result analysis and documentation.

11. Environmental Needs

11.1 Hardware Requirements

Component	Requirement
Computer/laptop	System capable of running Ubuntu and Mininet.
Processor	Multi-core processor recommended.
RAM	8 GB or more recommended.
Storage	Sufficient free space for source files, logs, and test results.
Network interface	Required for virtual network communication.

11.2 Software Requirements

Software/Tool	Purpose
Ubuntu Linux	Operating system for project execution.
Python 3	Runs client, server, controller, and test scripts.
Mininet	Creates the virtual network topology.
POX controller	Manages SDN policies.

Open vSwitch	Provides virtual network switching.
TCP socket programming	Provides client-server communication.
Terminal / code editor	Used to execute and edit project files.
Matplotlib and Pandas	Used for graph generation and CSV analysis.

11.3 Network Requirements

- Mininet topology with one server, ten client hosts, and one Open vSwitch.
- Server IP address: 10.0.0.100.
- Server port: 5000.
- TCP communication between clients and the server.
- OpenFlow communication between the POX controller and Open vSwitch.
- Controller socket path: /tmp/sdn_ctl.sock.

11.4 Test Data Requirements

- Valid and invalid user credentials.
- Sample files for upload and download tests.
- 5 MB and 20 MB files for performance tests.
- Multiple client hosts for concurrent transfer tests.

12. Responsibilities

All three members of Team 19 participate in project testing, verification, and documentation. Specific task ownership may be assigned by the team during execution.

Team Member	USN	Responsibility
SRUSHTI PURANIK	PES1UG25AM408	Participates in project development, functional testing, and documentation.
SHRITHL S SHETTY	PES1UG25AM383	Participates in network setup, SDN controller testing, and integration testing.
VAMSHIK KUDLUR M R	PES1UG25AM436	Participates in client-server testing, performance testing, result analysis, and documentation.

All team members contribute to test execution, identifying errors, verifying results, and preparing the final test report.

13. Staffing and Training Needs

13.1 Staffing Requirements

Testing is carried out by the three members of Team 19. No additional testing staff is required. Team members share responsibility for functional and integration testing, security checks, SDN policy verification, performance testing, and documentation.

13.2 Training Requirements

30. Python programming and TCP socket communication.
31. Linux commands and the Ubuntu environment.
32. Mininet network topology creation.
33. POX SDN controller and OpenFlow concepts.
34. File handling and client-server architecture.
35. Performance measurement and analysis using CSV files and Python graphs.

This knowledge is developed through project implementation, laboratory practice, and testing activities.

14. Schedule

The following is an indicative sequence for the testing activities. The actual dates may be adjusted to match the team's project timeline.

No.	Testing Activity	Planned Schedule
1	Test environment setup	Day 1
2	Mininet topology and controller testing	Day 2
3	Functional testing of login and file operations	Day 3
4	Security and permission testing	Day 4
5	SDN policy testing	Day 5
6	Concurrent file transfer testing	Day 6
7	Performance testing with SDN enabled and disabled	Day 7
8	Result analysis and graph generation	Day 8
9	Test documentation and final report preparation	Day 9

15. Risks and Contingencies

No.	Risk	Impact	Contingency Plan
1	SDN controller fails to start	Network policies may not work.	Restart the controller and verify switch connectivity.
2	Mininet topology creation fails	Testing cannot proceed.	Clean up the previous network and recreate the topology.
3	Server becomes unavailable	File transfer tests may fail.	Restart the server and verify that port 5000 is listening.
4	Incorrect credentials	Authentication tests may give unexpected results.	Verify configured usernames and passwords.
5	File transfer failure	Upload or download testing is interrupted.	Check server availability, client commands, and network connections.
6	Bandwidth variation	Performance results may vary between runs.	Repeat tests under similar conditions and compare measurements.
7	Insufficient system resources	Concurrent transfers may slow down or fail.	Reduce background processes and repeat tests.
8	Test data loss	Results may become unavailable.	Keep backups of CSV files, logs, and graphs.